

Sumanth Rao

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EDUCATION AND SKILLS

University of California, San Diego - Ph.D. in Computer Science 2021 - 2027 (expected)
Advisors: [Stefan Savage](#) and [Geoffrey M. Voelker](#)
PES University - B.Tech. in Computer Science, First Class with Honors 2016 - 2020

RESEARCH AND INDUSTRY EXPERIENCE

Google - Gmail Delivery Security | *Student Researcher* Jun 2026 - Sep 2026

- Built an observatory for domain hijacks used to bypass SPF, correlating domain and IP ownership signals.
- Deployed into Google prod, scanning 200M Gmail domains every 24 hours, with actionable alerts for < 0.1% of domains.

Corelight Labs | *Research Intern* Jun 2025 - Sep 2025

- Developed Zeek-based fingerprints for malicious Domain Controller activity in east/west enterprise traffic.
- Converted noisy production telemetry into detection signatures suitable for deployment across customer networks.

Google Cloud - Virtual Cluster | *Software Engineering Intern* Jun 2022 - Sep 2022

- Improved Borg cluster debugging by surfacing invocation history across Colossus, Spanner, and Persistent Disk.

Citrix - Licensing | *Software Engineer I* Aug 2020 - Aug 2021

- Built Docker-based hybrid-cloud test automation, reducing Citrix License Server test resource use by up to 50%.
- Earned the highest tier of Citrix's internal software security certification.

Zulip Open Source Project | *Founding contributor, Zulip Terminal; Google Summer of Code (GSoC)* 2019 - Present

- Helped launch the terminal client (GSoC 2021) and later mentored five GSoC students (2022-2026).

SELECTED SECURITY RESEARCH PUBLICATIONS

Lost in Translation: Text Message Spoofing via Email
IEEE S&P 2026 - Distinguished Paper Award • First author; disclosed [CVE-2025-46311](#); patched by Apple.

BLE Theft Auto: Aftermarket Automotive Remote Control Security
USENIX Security 2026 • Evaluated BLE remote-control systems; [CVE-2026-18411](#); Disclosure via CISA; *Wired*; DEFCON 34 talk.

Patient Care Technology Disruptions Associated With the CrowdStrike Outage
JAMA Network Open 2025 • Co-first author; measured disruptions across U.S. hospitals; covered by *Wired* and *Forbes*.

Unfiltered: Measuring Cloud-based Email Filtering Bypasses
ACM Web Conference 2024 • First author; large-scale study of email filtering bypasses; covered by *DarkReading*

IRRegularities in the Internet Routing Registry
ACM Internet Measurement Conference 2023 • Co-author; empirical study of Internet Routing Registry data.

No Privacy Among Spies: Insecurity of Consumer Android Spyware Apps
Privacy Enhancing Technologies Symposium 2023 • Co-author; presented at FTC PrivacyCon 2024.

TECHNICAL SKILLS

Languages C++, Python, Rust, Go, JavaScript, Shell, SQL.
Internet Measurement Censys, PassiveDNS, ZMap, RIPE Atlas, WHOIS, Nmap.
Network Analysis Zeek, Wireshark, mitmproxy, traffic fingerprinting.
Reverse Engineering Ghidra, JADX, Android app analysis.
Systems & Infrastructure Linux, Docker, Git, Django.

AWARDS, RECOGNITION, AND SERVICE

- Distinguished Paper Award**, IEEE S&P 2026.
- 2026 CSE Doctoral Award for Excellence in Service and Leadership;
- Program Committee**, RAID 2026.
- Provisional U.S. Patent 63/674,158, "Digital Availability Surveillance Method," assigned to UC San Diego (2024).
- Teaching assistant, UC San Diego: Intro to Computer Security and Graduate Operating Systems (2022-2023).
- CSE Visit Day Coordinator (2026); UCSD GradWIC Allyship Co-Chair (2023-2024); Jacobs Graduate Student Council VP External (2021-2023).